QRS Complex Detection in ECG Signal for Wearable Devices

M. Riadh Arefin, Kouhyar Tavakolian, *Member, IEEE* and Reza Fazel-Rezai*, IEEE Senior Member*

***Abstract*— This paper presents QRS complex detection algorithm based on dual slope technique, which is suitable for wearable electrocardiogram (ECG) applications. For cardiac patients of different arrhythmias, ECG signals are needed to be monitored over an extensive period of time. Thus, the wearable heart monitoring system needs computationally efficient QRS detection technique with good accuracy. In this paper, a method of QRS detection based on two slopes on both sides of an R peak is presented which is computationally efficient. Based on the slopes, first, a variable measuring steepness is developed, then by introducing an adjustable R-R interval based window and adaptive thresholding techniques, depending on the number of peaks detected in such window, R peaks are detected. The algorithm was evaluated against MIT/BIH arrhythmia database and achieved 99.16% detection rate with sensitivity of 0.9935 and positive predictivity of 0.9981. The method was compared with two widely used R peaks detection algorithms.**

***Index Terms* - Electrocardiogram, QRS detection, wearable sensors, low complexity.**

I. Introduction

As the population ages, the economic impact of cardiovascular diseases on health care system increases. Thus, there is a growth of interest in wearable technology to make health care delivery system more efficient. Therefore, physicians can use such systems to monitor several patients at a time and provide regular feedback reducing the number of visits. To obtain accurate response with less computational complexity as well as long battery life time in such monitoring system, there is a demand for developing fast and accurate algorithms and prototypes. A computationally efficient QRS detection algorithm is indispensable for low power operation on ECG signal.

In need of detecting QRS complex, within the last few decades many new algorithms have been proposed [1]. Algorithm based on derivatives, digital filtering, wavelet transforms, neural networks, filter banks, hidden Markov models, genetic algorithms as well as heuristic methods mostly based on nonlinear transformation are all developed to detect QRS complexes more effectively. But the evolution of these algorithms is driven by the performance of the detection not the computational complexity [1]. Hence most of the time, these methods are not suitable for giving a long battery performance in wearable devices. Wang *et al.* [2] proposed a novel dual slope QRS detection algorithm with less computational complexity as well as high accuracy. The algorithm is based on the fact that the largest change of slope usually happens at the peak of QRS complex considering the width of the complex is relatively fixed, in range of 0.06-0.1 seconds [3]. However, a time consuming slope calculation for each sample has been made to get high variable values in the QRS complex. Such slope calculation can be avoided by highlighting only one sample from each side and taking advantage of the multiplication of slopes. In this paper, a new dual-slope based QRS complex detection algorithm is presented by introducing an adjustable window based on the R-R interval and adaptive thersholding techniques based on the number of peaks detected in such a window. The proposed method is fast with high accuracy.

*Research was supported by ND NASA EPSCoR.

M. R. Arefin is with the Biomedical Research Complex (BERC), Electrical Engineering Department, University of North Dakota, Grand Fork, ND 58202 USA (email: mriadh.arefin@my.und.edu).

K. Tavakolian is with Biomedical Research Complex (BERC), Electrical Engineering Department, University of North Dakota, Grand Forks, ND 58202 USA (email: kouhyar@und.edu).

R. Fazel-Rezai is with Biomedical Research Complex (BERC), Electrical Engineering Department, University of North Dakota, Grand Forks, ND 58202 USA (email: reza@und.edu, phone: 701-777-3368, fax: 701-777-5253).

II. Methodology

Typically, the Q, R and S peaks are three deflections which depict a single event and occur in a periodic manner in ECG signal. Starting from Q wave, a downward deflection, R wave follows with steepest upward deflection and S wave is any downward deflection after the R wave. The time taken by this event is relatively fixed, in the range of 0.06-0.1 seconds. If we calculate the slope of straight line between two samples, which is about, half the width of QRS complex away from each other, the largest values of slope should be found in QRS complex. Moreover, QRS complexes occur with a certain interval between them, normally in the range of 0.6-1 seconds [4]. Therefore, by introducing a window of time which is automatically adjustable to ECG changes such as QRS morphology and heart rate, a faster QRS detection process is possible. Within this window, adaptive thersholding techniques can be used in order to consider a signal section as QRS complex. If thersholding criteria are satisfied then local extreme can be searched within QRS complex to locate R peak. Based on this idea, this new QRS detection algorithm is proposed.

A. Calculation of slopes

As half of QRS complex width is in the range of 0.03-0.05 seconds, the processing of sample begins by calculating one slope between the current sample and the sample 0.3 seconds away on both sides. From the range of 0.3-0.5 second, this distance between samples is carefully chosen to 0.3 seconds over others in order to eliminate the possibility of low slope values caused by the different shapes of QRS complexes. The equations to calculate the slopes are:

$$S_{Left} = \left(\frac{ECG(i) - ECG(i-k)}{k}\right) \quad (1)$$

$$S_{Right} = \left(\frac{ECG(i) - ECG(i+k)}{-k}\right) \quad (2)$$

where $ECG(i)$ is the current sample. $ECG(i-k)$ and $ECG(i+k)$ are the samples which are $0.03f_s$ away from both sides of the current sample. f_s is the sampling frequency and k is equivalent to the nearest integer value of $0.03f_s$. By multiplying S_{Left} with S_{Right} for each sample, a steepness measuring variable S_{Mult} is evaluated as follows:

$$S_{Mult} = S_{Left} \times S_{Right} \quad (3)$$

As the sign of slope on both sides should be opposite at R peaks, the following condition is verified:

$$sgn(S_{Left}) = -sgn(S_{Right}) \quad (4)$$

If the sign of slopes on both sides is opposite, this steepness measuring variable S_{Mult} indicates very high values of this variable around QRS complex as shown in the Fig. 1.

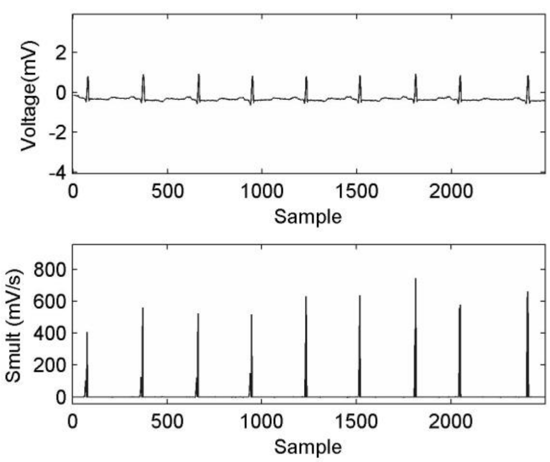

Figure 1. Tape 100 of MIT-BIH database shohwing values of variable $S_{Mult.}$

To eliminate false beat detection, the S_{Mult} must be larger than a threshold value. In order to define a universal adaptive thresholding technique, a learning stage based on the first 3 seconds of data is introduced where at least 2 QRS complexes are required to occur to initialize parameters based on RR-interval. Therefore, to initialize detection threshold and RR-interval based parameters, at first S_{Mult} has been calculated for the first 3 seconds. Then, the threshold is considered as:

$$T = \frac{Max(S_{mult})}{3} \quad (5)$$

This means a signal is considered as R peaks if the value of S_{Mult} is greater than one-third of the maximum value of S_{Mult} for the first 3 seconds. In order to adapt with the ECG changes, for subsequent signal sections, the threshold was updated after every 8 detected peaks based on the average value of S_{Mult} of these peaks. The updating equation is:

$$T = avg(S_{Mult_R}) \quad (6)$$

where S_{Mult_R} is the average value of S_{Mult} variable of last eight detected peaks.

B. R-R interval window

An R-R interval is the interval between two R peaks which is in the range of 0.6-1 seconds for normal resting adult human heart. But R peaks can occur faster than the average to respond to external stress. However, once an R peak is detected, there is a 200 milliseconds refractory period before the next R peak could since physiologically occur [4]. For faster detection of QRS complex, a "moving window" of time can be used based on most recent RR interval values. To achieve reliable performance, this moving window of time is calculated as follows:

$$RR_{int} \pm \Delta R \quad (7)$$

where RR_{int} is the average interval between last 8 detected peaks and ΔR is considered as $0.48f_s$. This ΔR is carefully chosen based on the refractory time and average RR-interval of human heart to eliminate the chances of creating false negatives.

In this period of time, for normal resting heart only one peak should be detected. But as mentioned earlier, R peaks can occur more quickly than the average. At the same time, T wave, P wave or noise sections can produce large slopes as well. However the height of this slopes are usually much smaller than those of QRS complex. To avoid wrong detection of R peaks, if two peaks are detected within single period of window, then both of them are compared with the average height of previously detected eight peaks H_{ave}. If they are larger than a certain threshold, then both of them are considered as R peaks. Otherwise the peak with high value of S_{Mult} is considered as R peaks only. Following is the condition:

$$H_{cur} > H_{ave} \times 0.7 \quad (8)$$

Within one period of window more than two peaks cannot be possible as based on recent RR_{int} the moving window is adapting with time. So, if the detection of the peaks is more than two then the peaks with lower values of S_{Mult} among them are ignored.

C. Search back technique

Sometimes the amplitude of the ECG as well as the amplitude of R peaks gets significantly shorter when switching from one patient's ECG to another. So it might be possible that peaks cannot be detected in that region because of low value of the calculated slopes while the value of threshold based on previously detected peaks is high. Such a case is shown in Fig. 2.

The algorithm needs to be rapidly adaptable without requiring any special learning phases in a situation like this. Thus for threshold re-adjustment, if in a period of time corresponding to average QRS complex interval elapses without a beat detection, the threshold is lowered by half of the previous threshold value. By searching back through the same time interval, the sensitivity of detection can be increased to avoid missing valid R peaks on that region.

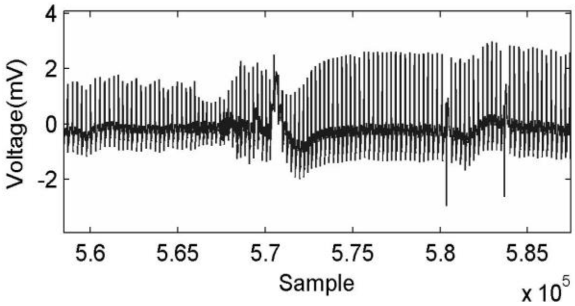

Figure 2. Tape 109 of MIT-BIH database showing variation of amplitude in ECG signal.

D. Adjustment within QRS complex

If all the above criteria are met then local maximum and minimum are searched in current signal section to determine the location of R peaks. To avoid multiple detections within one QRS complex, the one with the largest value of S_{Mult} is retained as an R peak, while the other is discarded.

III. Evaluation and Comparison

Several QRS detection algorithms have been proposed based on software during last 40 years. However, this large variety of QRS detection algorithms and continuous attempt to enhancement indicates that no single algorithm is clearly superior over all others for detection. This is due to large diversity of QRS waveform and wide variation of noise and artifact in ECG signal [1]. Among all these algorithms, Pan and Tompkins algorithm [4] is the most widely used and often cited for QRS detection. On the other hand, from the literature [1], algorithm developed by Afonso *et al.* [5] has lower computational load as well as good accuracy.

As the cornerstone of an algorithm evaluation is to compare it with others, because of the high acceptability of Pan and Tompkins method and low computational complexity of Afonso method, these two algorithms are used to compare with this new approach.

As both of the algorithms are evaluated on MIT-BIH database [6] by the respective authors, all the algorithms are evaluated using this database. It is a standard database of two channel 48 half-hour ECG recordings. These recordings have 11 bit resolution over 10 mV and are sampled at 360Hz. Testing was performed on channel 1.

To analyze the performance, true positive (TP), false negative (FN) and false positive (FP) were evaluated. When an algorithm detects R peaks correctly a true positive happens. A false negative (FN) occurs when algorithm fails to detect an actual QRS complex quoted in the corresponding annotation file of the record. And false positive (FP) means a false beat detection. In order to compare these algorithms three benchmark parameters are used. Using FP and FN, error rate (ER) is calculated based on the following equation:

$$ER(\%) = \frac{FP + FN}{Total\ QRS} \tag{9}$$

where total QRS is total number of QRS complex in the ECG data.

The other two parameters, sensitivity and positive predictivity, are computed by [5]

$$Se = \frac{TP}{TP + FN} \tag{10}$$

$$+P = \frac{TP}{TP + FP} \tag{11}$$

The percentage of true beats that were detected correctly is reported by sensitivity *Se*. The positive predictivity $+P$ reports the percentage of beat detection which are true beats according to annotation.

IV. Results

Since Tompkins method and Afonso method were not applied to the same length of data, in order to get an impartial comparison of the accuracy and complexity, all three algorithms were evaluated in same length of data using MATLAB environment on the same computer. The MATLAB code for Tompkins's approach is provided by [7]. For Afonso technique, code from [8] has been used. Table I shows the accuracy of these codes according to their error rates mentioned by the authors on MIT-BIH database. The accuracy of these m files (MATLAB files) is close enough to consider them for evaluation of these techniques.

TABLE I. ACCUARACY OF THE M FILES

Method	Total QRS	Paper			M file		
		FP	*FN*	*Error rate (%)*	*FP*	*FN*	*Error rate (%)*
Pan and Tompkins [4]	109,508	507	277	0.72	441	757	1.09
Afonso [5]	90,782	406	374	0.86	1,189	613	1.98

Table II shows the summary of QRS detection for all algorithms based on FN and FP. T is the runtime of the corresponding program using the same MATLAB environment on the same computer for all methods. As mentioned in [1], Tape 108 is very noisy and has very low SNR. This newly developed algorithm has 41 FP's and 41 FN's whereas the other algorithms have more than 100 false detections. As the variable S_{Mult} has been calculated based on the width of QRS complex, this algorithm is sensitive to the width. In this database, some of the annotated R peaks because of premature ventricular contraction (PVC) have more width than the usual range. Tape 208 has more than 992 beats with wider width. Because of this reason the sensitivity of the algorithm (detection of FN's) is greatly affected by PVC. Those PVCs have been eliminated before running the algorithm. The total number of deleted signal contains 1,041 R peaks and is 0.067% of the total data. Tape 208 was not considered because of its high PVC contains.

TABLE II. PERFORMANCE OF THE ALGORITHMS USING MIT/BIH DATABASE

Tape	Total	Pan Tompkins [4]			Afonso [5]			Proposed Dual-slope based Algorithm		
		FN	*FP*	*T (sec)*	*FN*	*FP*	*T(sec)*	*FN*	*FP*	*T(sec)*
100	2273	0	0	5.23	2	0	2.10	2	0	2.45
101	1865	2	3	3.72	1	2	1.67	1	2	2.13
102	2187	0	0	3.93	1	0	1.69	4	0	2.26
103	2084	0	0	3.41	2	0	1.68	2	0	2.33
104	2229	104	101	3.79	44	81	1.72	16	49	2.36
105	2572	17	43	4.22	11	40	1.74	5	23	2.66
106	1507	0	0	3.81	0	0	1.75	0	0	1.98
107	2137	18	2	3.54	20	58	1.73	16	1	2.43
108	1774	125	282	4.54	19	184	1.68	41	41	2.05
109	2532	5	0	2.81	97	3	1.72	15	0	2.71
111	2124	9	0	3.88	3	2	1.68	1	0	2.38
112	2539	0	0	3.52	2	1	1.72	1	0	2.66
113	1795	0	0	3.68	3	0	1.67	1	0	2.07
114	1879	2	7	4.30	2	4	1.69	23	6	2.03
115	1953	0	0	3.91	3	0	1.68	1	0	2.20
116	2412	24	4	3.41	24	2	1.72	27	0	2.46
117	1535	0	0	3.88	2	0	1.70	1	0	1.80
118	2278	0	1	3.82	2	4	1.69	1	0	2.50
119	1543	0	1	3.86	0	0	1.70	0	0	1.07
121	1863	2	0	3.91	2	3	1.68	3	0	2.19
122	2476	0	0	3.44	4	0	1.68	1	0	2.65
123	1518	3	0	3.92	2	0	1.63	4	0	1.80
124	1572	0	0	3.37	0	1	1.64	0	0	1.90
200	2601	4	1	4.05	8	14	1.74	9	7	2.23
201	1625	0	0	3.95	0	0	1.69	0	0	1.91
202	2136	8	1	3.63	10	0	1.69	46	0	2.12
203	2980	4	19	3.98	60	51	1.80	212	34	2.41
205	2656	8	0	3.78	10	0	1.75	49	0	2.68
207	1477	0	4	3.77	3	16	1.70	8	5	2.87
209	3004	0	1	4.19	2	2	1.78	2	0	2.86
210	2423	0	5	3.74	0	9	1.75	0	5	2.52
212	2748	0	1	3.92	3	2	1.75	1	0	2.82
213	2641	0	0	4.17	0	0	1.90	0	0	3.03
214	2265	3	0	3.72	10	2	1.71	19	0	2.32
215	2533	0	0	5.12	0	0	1.82	0	0	2.87
217	2209	8	1	3.42	4	2	1.71	10	1	2.42
219	1553	0	0	3.48	0	0	1.73	0	0	2.32
220	2048	0	0	3.70	3	0	1.65	2	0	2.28
221	2031	0	0	3.85	0	0	1.75	0	0	2.18
222	2483	18	19	4.38	3	2	1.75	31	1	2.27
223	2605	10	0	3.75	9	0	1.74	59	0	2.60
228	2053	16	7	4.32	7	61	1.70	18	17	2.17
230	2256	0	0	4.30	3	2	1.68	2	0	2.45
231	1571	0	0	3.75	2	0	1.77	2	0	2.05
232	1780	1	2	4.66	1	9	1.76	2	1	2.03
233	3079	2	0	3.85	22	1	1.82	18	0	2.53
234	2753	1	0	4.00	6	0	1.75	5	0	2.84
Total	**102157**	**394**	**505**	**3.90**	**412**	**558**	**1.73**	**661**	**193**	**2.34**

Furthermore, Table III shows the overall error rate, sensitivity, positive predictivity and processing time based on TP, FP and FN. As shown in the table, Because of the higher positive predictivity the overall error rate is higher. Due to wider width of PVCs the sensitivity of this method is lower than the others. On the other hand, the complexity has been analyzed based on time taken to run the code. The proposed method is faster than Pan Tompkins by 1.56 seconds and slower than Afonso's method by 0.61 seconds which can be considered small enough for wearable devices.

TABLE III. COMPARISON OF ALGORITHMS

Method	Total TP	FN	FP	Accuracy	Sensitivity	Positive Predictivity	Proces-sing Time
Afonso [5]	101745	412	558	99.0505	0.9960	0.9945	1.73
Pan Tompkins [4]	101763	394	505	99.1200	0.9961	0.9951	3.90
Proposed Algorithm	101496	661	193	99.1640	0.9935	0.9981	2.34

V. CONCLUSION

In this paper, a new QRS detection algorithm was presented and its performance was compared with the performance of Pan and Tompkins's and performance of Afonso's algorithm when applied to MIT-BIH database. With comparatively low computational complexity, the overall accuracy of the method is higher than the other two algorithms. Hence the method is highly suitable for wearable devices.

REFERENCES

[1] B. U. Kohler, C. Hennig, R. Orglmeister, "The principles of software QRS detection," *IEEE Engineering in Medicine and Biology Magazine,* vol. 21, no. 1, pp. 42,57, Jan.-Feb. 2002.

[2] Y. Wang, C. J. Deepu and Y. Lian, "A computationally efficient QRS detection algorithm for wearable ECG sensors," in *Conf. Rec. 2011 Annu. Int. Conf. Engineering in Medicine and Biology Society,* pp. 5641–5644.

[3] F. G. Yanowitz. (2006, 5 Dec). *Characteristics of the Normal ECG* [Online]. Available: http://ecg.utah.edu/

[4] J. Pan and W. J. Tompkins, "A Real-Time QRS Detection Algorithm," *IEEE Trans. Biomedical Engineering,* vol. BME-32, no. 3, pp. 230–236, March 1985.

[5] V. X. Afonso, W. J. Tompkins, T. Q. Nguyen, and S. Luo, "ECG beat detection using filter banks," *IEEE Trans. Biomed. Eng.*, vol. 46, pp. 192-202, 1999.

[6] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. Ch. Ivanov, R. G. Mark, J.E. Mietus, G.B. Moody, C. K. Peng and H. E. Stanley. (2000, 13June). *PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals[Online].* Available: http://circ.ahajournals.org/cgi/content/full/101/23/e215.

[7] Complete Pan Tompkins Implementation ECG QRS detector [Online]. Available: http://www.mathworks.com/matlabcentral/fileexchange/45840-complete-pan-tompkins-implementation-ecg-qrs-detector

[8] EEGLAB [Online]. Available: http://sccn.ucsd.edu/eeglab/